

Programming Exam 8

Date/Time: 2026.06.02 08:00 – 08:50

(程式檔命名學號_pe8.py，上傳至 Moodle PE8 上傳區)

Problem: Data Analysis on NBA 2022-23 Standings

Write a Python code to read NBA 2022-23 standings from a CSV file named “pe8_data.csv”. The CSV file contains multiple columns, and the first 9 columns with their meanings in order are: Conference (Eastern or Western), Team, W-L (Win-Loss Record), PCT (win-loss percentage), GB (game back, how far behind the team is from the leading team), PF (points for), PA (points against), Home (home record), Away (away record). Based on this file, answer the following questions.

- (1) Parse W-L and CONF records to compute each team's out-of-conference win percentage (i.e., wins vs. the other conference) and list the five teams with the highest out-of-conference rates. (解析 W-L 與 CONF 欄位，計算每支球隊的跨區(東區或西區)勝率，也就是對另一區球隊的勝率，並列出跨區勝率最高的前 5 支球隊。)
- (2) Parse HOME and AWAY records into separate win percentages, then find all teams that rank in the top third of their conference for both home and away win rates. (解析 HOME 與 AWAY 欄位為單獨的主/客場勝率，並找出在所屬西區或東區中同時位於主場及客場勝率前三分之一的球隊。)
- (3) Calculate each team's Pythagorean expected win percentage using $PF^2 / (PF^2 + PA^2)$, estimate expected wins ($\times 82$ games), and list the five teams that most overperform and underperform this expectation. (使用公式 $PF^2 / (PF^2 + PA^2)$ 計算每支球隊的畢達哥拉斯預期勝率，再乘以 82 場得出預期勝場，並列出超越與低於預期最多的各五支球隊。)
- (4) Find a partition of all 30 teams into two groups whose total PF sums differ by as little as possible—i.e. solve the 3-way partition problem—and report the three groups and their PF sums. (將 30 支球隊分成三組，使三組的 PF 總和差異最小，給出分組及各組 PF 總和。)
- (5) Pair up all 30 teams into 15 disjoint “games” (each team appears exactly once). For each complete pairing, compute the sum of absolute differences in PCT between paired teams, and find the pairing that **minimizes** that total. (將 30 支球隊配成 15 場「對戰」，每隊只出現一次。對每一種完整配對計算配對之間 PCT 絕對差的總和，並找出能使該總和**最小化**的配對方案。)
- (6) Find the longest list of teams where each team's win percentage is within 1% of the next one's. List that chain. (找出一條最長的球隊清單，使得每支球隊的勝率與下一支相差不超過 1%。列出該連續隊列。)

No Sample Input and Output for this PE. By running your program, make sure the answer to each question is clearly presented. You can decide the output format by yourself.

重要規定：從此次 PE7 開始，允許同學們使用生成式 AI 工具如 ChatGPT 來協助作答。請務必使用**自己能夠理解的 Python 程式語法來作答**。若我們認為你使用超過考試範圍內教的語法以外的進階或特殊語法，將會請你前來說明解釋該語法如何運作。若無法清楚說明該語法如何輔助解出此題，將視狀況對分數折扣或不予計分。

Note: You need to write comments (註解) for each part in your code.